



Medical Microbiology

Lecture Twelve

Enterobacteriaceae

Third stage- College of Dentistry

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Introduction:

- **Enterobacteriaceae** are a large family of Gram-negative bacteria that includes a number of pathogens such as *Klebsiella*, *Enterobacter*, *Citrobacter*, *Salmonella*, *Escherichia coli*, *Shigella*, *Proteus*, *Serratia* and other species.
- These pathogens are present in the human intestinal tract and are a normal part of the gut flora.

- They are a common cause of urinary tract infections (UTIs), and some species can also cause diarrhea.
- These pathogens can spread to the bloodstream resulting in life-threatening complications.
- Differentiation is based on biochemical reactions and differences in antigenic structure.

Family Enterobacteriaceae

Primary pathogens

Bacteria capable of causing disease in anyone

Shigella

Salmonella

Yersinia

Escherichia coli

Klebsiella pneumoniae

Opportunistic pathogens

Bacteria that can only cause disease under certain conditions or certain hosts

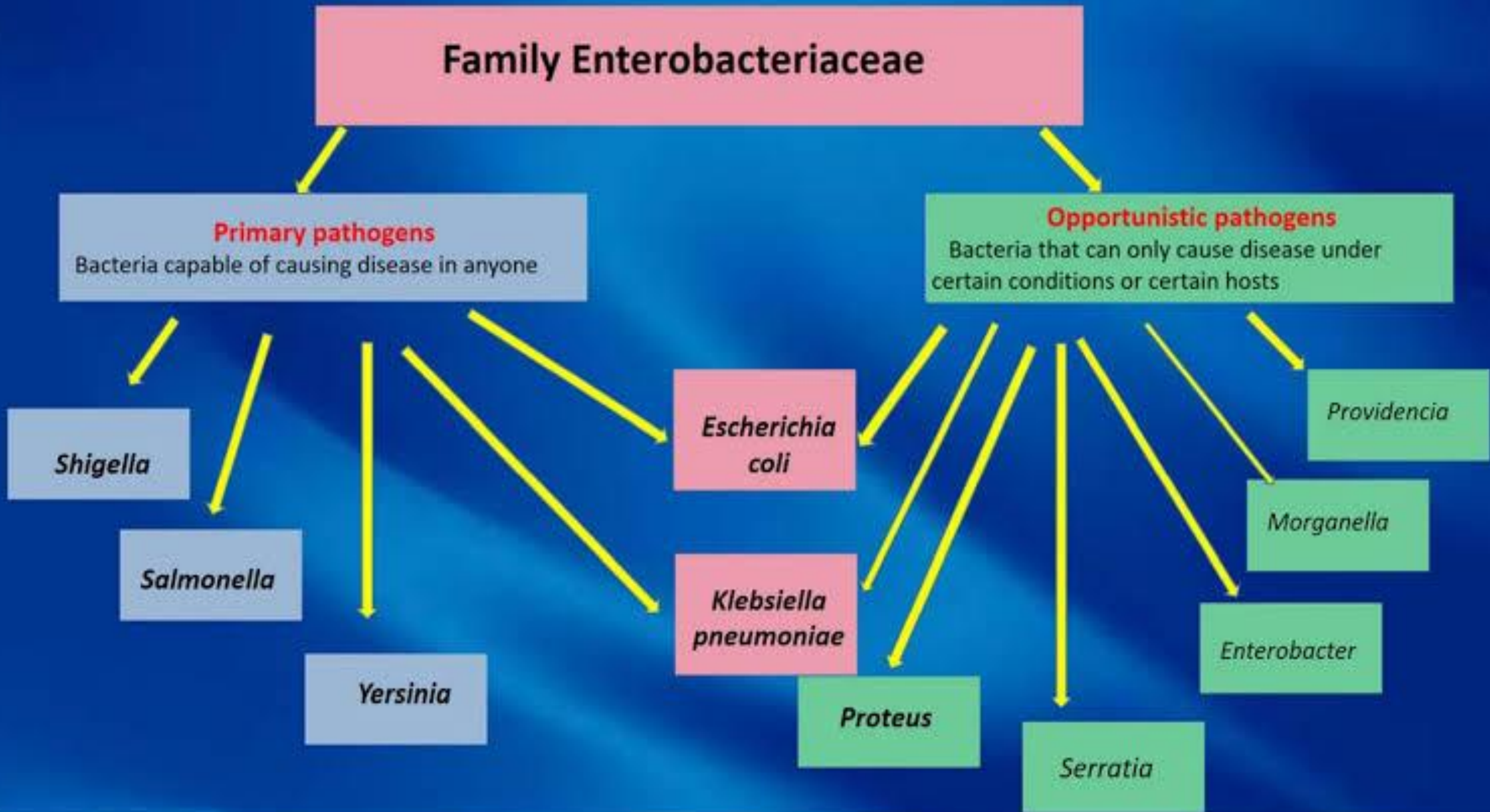
Providencia

Morganella

Enterobacter

Proteus

Serratia



General Characteristics of Enterobacteriaceae:

- 1- Gram-negative rods,
(growth on macconkey agar).
- 2- Facultative anaerobes.
- 3- Non-spore-forming.
- 4- Oxidase-negative (except some species).
- 5- Catalase-positive.
- 6- Ferment glucose with acid and gas production.
- 7- Reduce nitrates to nitrites.
- 8- Motile or non-motile (depending on species).



Classification of Enterobacteriaceae:

Enterobacteriaceae can be categorized based on their clinical relevance:

1. Commensal Enterobacteriaceae:

- *E. coli* (beneficial strains in gut micro-biota).
- *Citrobacter* (rarely pathogenic).

2. Opportunistic Enterobacteriaceae:

- *Escherichia coli* (UTIs, sepsis, gastroenteritis).
- *Klebsiella* (pneumonia, bloodstream infections).
- *Enterobacter* (nosocomial infections).
- *Proteus* (UTIs, wound infections).
- *Pseudomonas* (Wound and burn infections, septicemia).

3. Pathogenic Enterobacteriaceae:

- *Salmonella* (causes typhoid fever, gastroenteritis).
- *Shigella* (causes bacillary dysentery).
- *Yersinia* (causes plague, enterocolitis).
- **Vibrio cholerae:** Cholera is an intestinal infection that can lead to severe, life-threatening watery diarrhea and vomiting.

Virulence Factors and Pathogenicity:

- 1- Endotoxin (LPS):** Causes fever, shock, and inflammation.
- 2- Capsule:** Some species possesses capsule that protects against phagocytosis like: (*Klebsiella pneumonia*).
- 3- Fimbriae (Pili):** Helps in adhesion to host tissues such as, (*E. coli*, *Salmonella*).
- 4- Exotoxins:** Enterotoxins and cytotoxins contribute to disease (Shiga toxin in *Shigella* and O157:H7 in *E. coli*).
- 5- Iron acquisition system:** Allow bacteria survival in iron-limited environments.

Clinical Importance in Dentistry:

Though primarily associated Enterobacteriaceae with gastrointestinal and systemic infections, they are also relevant to oral and dental health:

1- Oral infections: Opportunistic species (Klebsiella, Enterobacter) can colonize the oral cavity, especially in immunocompromised patients.

2- Periodontal disease: Some strains of E. coli and Klebsiella have been found in periodontal pockets.

3- Nosocomial infections in dental clinics: Contaminated waterlines or equipment may harbor Enterobacteriaceae, posing a risk to patients.

Laboratory diagnosis and Identification of Enterobacteriaceae:

Enterobacteriaceae are identified in the laboratory by using:

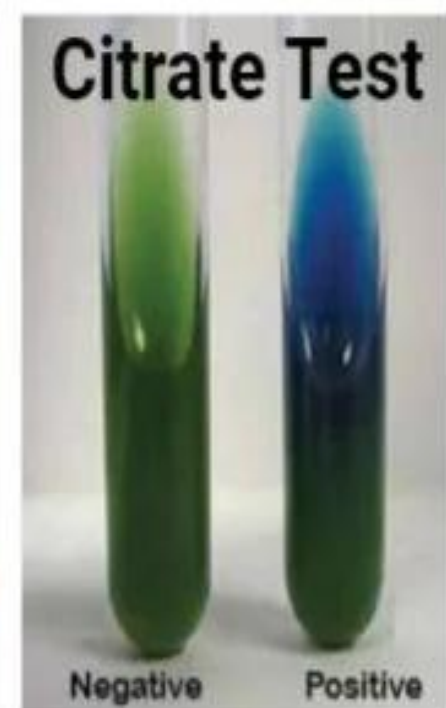
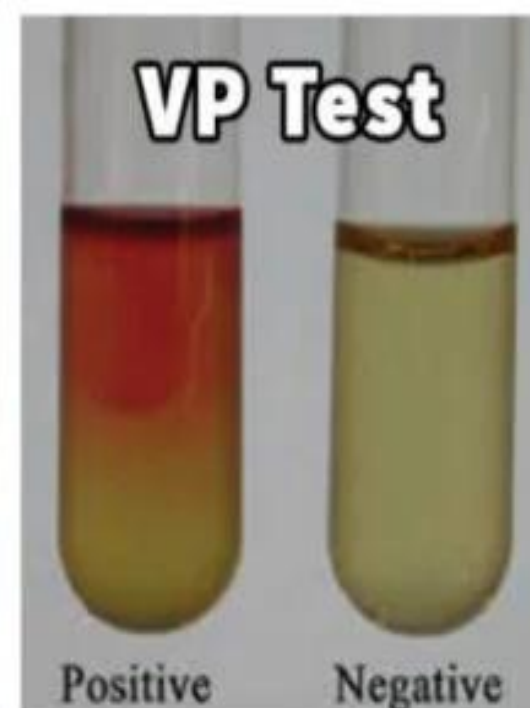
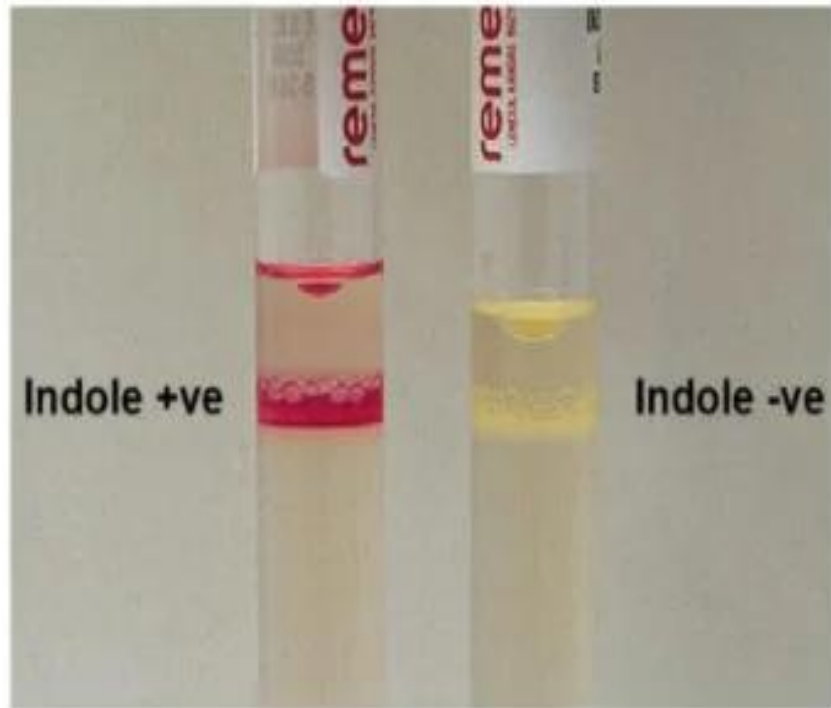
1- Gram Staining : Gram-negative rods.

2- Culture media: Growth on MacConkey agar some species (lactose fermentation).

3- Biochemical Tests:

- **IMVC** (Indole, Methyl Red, Voges-Proskauer, Citrate), **Urease**, **TSI** (Triple Sugar Iron that contain Glucose, Lactose, Sucrose, phenol red, in addition to Iron salt).
- **Oxidase Test:** Negative except: Pseudomonas aeruginosa, Vibrio cholerae, Neisseria spp. and Campylobacter spp.
- **Motility Test:** Variable (some are motile, some are not).
- If motile, motility is by **peritrichous flagella**.

IMViC Tests

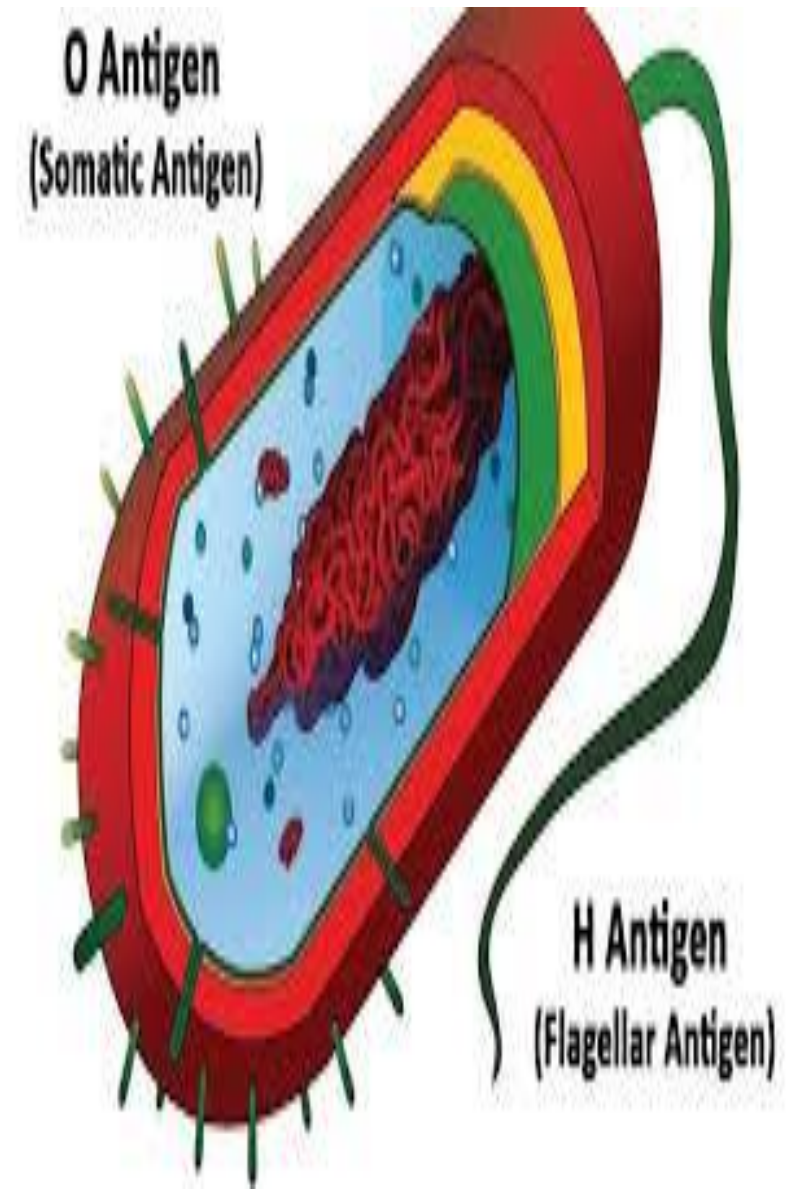


4- Serological Tests: for detecting specific surface antigens to identify bacterial species and strains.

The main antigens are:

- **O Antigen:** Part of the lipopolysaccharide (LPS) in the outer membrane; used for serotyping for detection bacteria (e.g., E. coli O157).
- **H Antigen:** Found on flagella; used to classify motile or non motile bacteria.
- **K Antigen:** Capsular antigen; helps bacteria evade the immune system.

5- Molecular Methods: PCR for precise identification.



Antibiotic Resistance in Enterobacteriaceae:

Many Enterobacteriaceae show increasing resistance, especially in hospital settings.

1-Major mechanisms:

- **Extended-Spectrum Beta-Lactamases (ESBLs):** Inactivate extended-spectrum beta-lactam antibiotics.
- **Carbapenem-Resistant Enterobacteriaceae (CRE):** Resistant to carbapenems (e.g., meropenem, imipenem), limiting treatment options.

2-Treatment (based on susceptibility testing):

- Cephalosporin
- Fluor quinolones (ciprofloxacin)
- Aminoglycosides (gentamicin)
- Colistin for multidrug-resistant strains

Prevention and Control in Dental Practice:

- 1- Strict infection control measures:** Sterilization of instruments, hand hygiene, and surface disinfection.
- 2- Proper use of antibiotics:** Avoid over prescription of drugs to prevent resistance.
- 3- Monitoring water quality:** Ensure dental unit waterlines are free from contamination.
- 4- Education and awareness:** Dental professionals should be aware of the risks associated with Enterobacteriaceae in oral and systemic infections.

Species	Lactose	H ₂ S	Motility	Urease	Notes
E. coli	+	–	+	–	cause of UTI, diarrhea
Klebsiella	+	–	–	+ (weak)	Capsule, pneumonia
Enterobacter	+	–	+	–	Opportunistic, hospital infections
Salmonella	–	+	+	–	Typhoid, gastroenteritis
Shigella	–	–	–	–	Dysentery, non- motile
Proteus	–	+	+ (swarming)	+	Strong urease, kidney stones
Yersinia	–	–	– (at 37°C)	+	Plague, mesenteric lymphadenitis

Question/ A stool sample from a patient with acute gastroenteritis was cultured. The organism showed the following results:

- Gram-negative rod
- Oxidase negative
- Non-lactose fermenter on MacConkey agar
- TSI: K/A with black precipitate
- Motile
- Urease negative

❖ **What is the most likely organism?**

- A) *Escherichia coli*
- B) *Shigella* species
- C) *Salmonella* species
- D) *Klebsiella pneumoniae*
- E) *Proteus mirabilis*

Correct Answer: C) Salmonella species

Explanation answer:

- Non-lactose fermenter → E. coli ,Klebsiella (تستبعد)
- K/A → glucose fermented only.
- Black precipitate → It means H_2S positive.
- Motile → Shigella(non motile).
- Urease negative → Proteus Urease=(strong +ve).
- (Salmonella) ✓ المواصفات تنطبق على



Any Question?



Thank you